STAGE 1 · V-PCC OVER QUIC

STAGE 2 · EDGE-TO-HMD

CLOUD · ORIGIN

1

Volumetric server

Caddy · HTTP/3 origin
V-PCC GOF chunks
TLS 1.3 · AES-128-GCM

Dynamic point cloud sequences: Soldier, Longdress, ThaiDancer (8iVFB, vox10/12), TMC2 v25 + x265.

QUIC

PRIVATE 5G SA (ERICSSON) · COLOCATED EDGE

impairment injected on the origin→edge path; renderer sits with the gNodeB

2

Impairment

Linux TC: TBF + netem
one UDP port per slice
loss · added latency

3

Edge renderer

Unity 6 · V3C decoder
RTX 4070 · ALVR server
per-GOF telemetry → ψ

ALVR

gNodeB

4

NC
BBU
RU

5G NR

A	**Clean**	no loss, no delay	**RA · GOF 32**
B	**Loss**	loss only	**LDP · GOF 4**
C	**Latency**	delay only	**LDP · GOF 4**
D	**Loss + latency**	both	**LDP · GOF 4**

Slices
2×2 factorial on B, C, D at a fixed encoder. A anchors the clean ceiling (Slice A).

USER END

5

Meta Quest 3

5G USB dongle
ACR rating (P.910)
no Wi-Fi last mile

Stage 2 is identical in every condition, so any MOS difference originates in the Stage 1 path.